Name:- Bhavgavi
Regno:- 192312147

Safe Drinking water Quality Screening using Concept learning and Decision trees.

1. Problem Statement:-

The objective is to develop an educational machine-learning system that screens a water sample and classifies it into:

→ Safe for use
→ Needs Treatment

The system uses simple categorical water quality attributes such as pH band, turbidity, TDS band residual chlorine. microbial indicator and source type.

The system demonstrates two approaches:-

1. Concept learning using Candidate Elimination / version space
2. Decision Tree learning using Entropy and Information Gain

1.2 learning problem:-

Let an instance be represented as:-

$x = (pH, Turbidity, TDS, Chlorine, Microbial, Source)$

The target concept is

$C(x) =$ Safe for use — If sample satisfies the learned conditions
Needs treatment — otherwise

1.3 Attributes

Attributes	Possible Values
PH Band	Acidic, Vestical, Alkaline
Turbidity	low, medium, high
TDS Band	low, medium, high
Residual chlorine	low, Normal, high
Microbial Indicator	Absent, present
Source Type	municipal, Ground water, Borewell, Surface

1.4 Classlabels

There are two target classes:

→ Safe for use

→ Need treatment

1.5 Hypothesis Representation:

We use a Conjunctive hypothesis representation Each hypothesis Contain either:

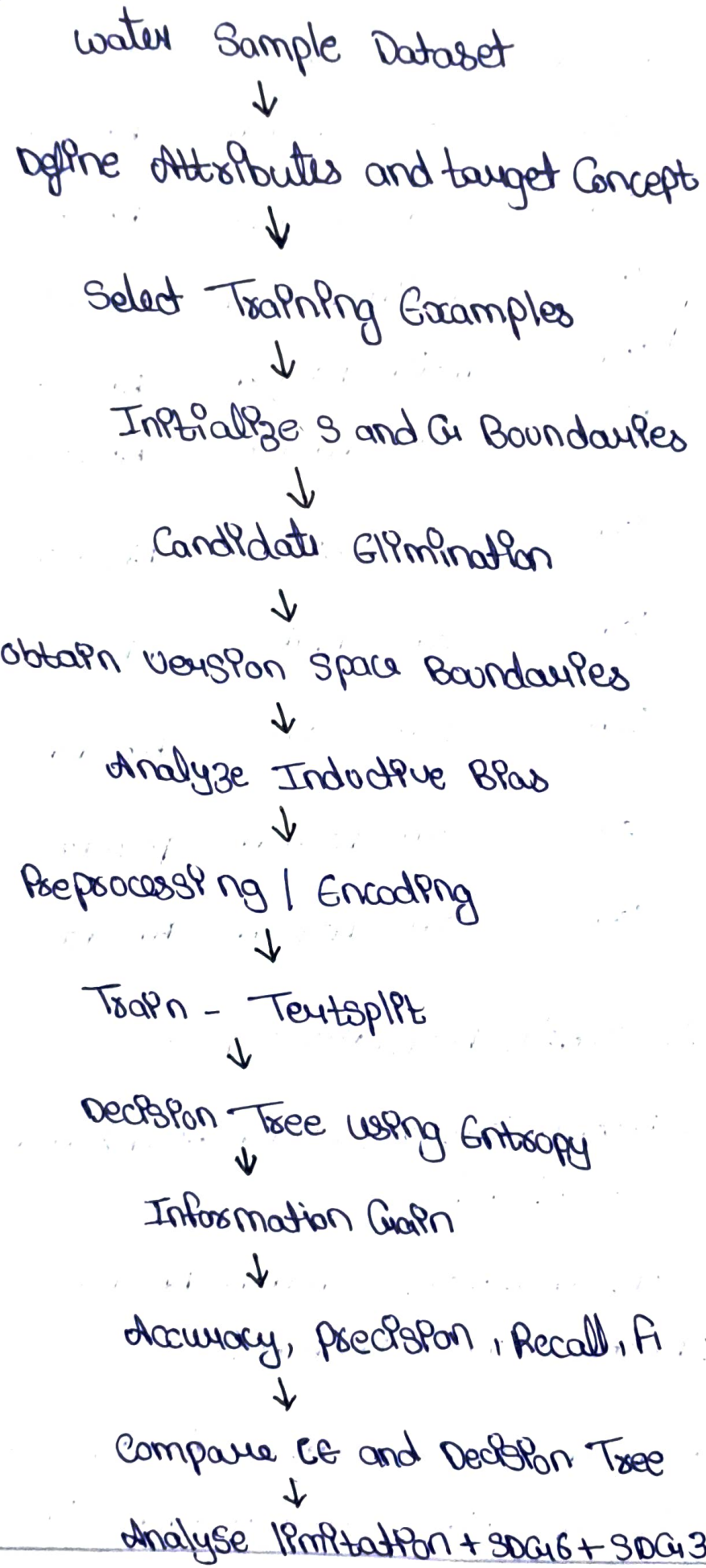
→ a specific attributes value or

→ ?, meaning any value is acceptable

for example:-

h = (Neutral, low, ?, Normal, Absent, ?)

2. learning workflow



3. Candidate elimination/version space.

S.No	PH	Tasbeclity	TDS	Chlorine	mpcsalpal	class.
1.	Neutral	low	low	Normal	Absent	Safe
2.	Neutral	low	low	normal	Absent	Safe
3.	Neutral	low	medium	normal	Absent	Safe
4.	Neutral	low	medium	normal	Absent	Safe
5.	Neutral	high	low	normal	Absent	need treatment
6.	Alkaline	low	low	normal	Absent	need treatment
7.	neutral	low	low	high	Absent	need treatment
8.	neutral	low	low	low	Absent	need treatment
9.	neutral	low	low	normal	present	need treatment
10.	neutral	low	low	normal	Absent	Safe.

Candidate Elimination Boundary Table:-

Positive → <Neutral, low, low, normal, Absent, manicipal> →
<?, ?, ?, ?, ?, ?>

Negative → <neutral, low, ?, normal, Absent, ?> →
<neutral, low?, normal, ?>

Positive <neutral, low ?, Normal, absent, ?> →
<neutral, low, ?, Normal, Absent, ?>

Conductive Bias

Inductive Bias is the set of assumption a machine-learning algorithm uses to generalize for limited training examples to unseen examples.

Example:- <Neutral, low, ?, Normal, Absent, ?>

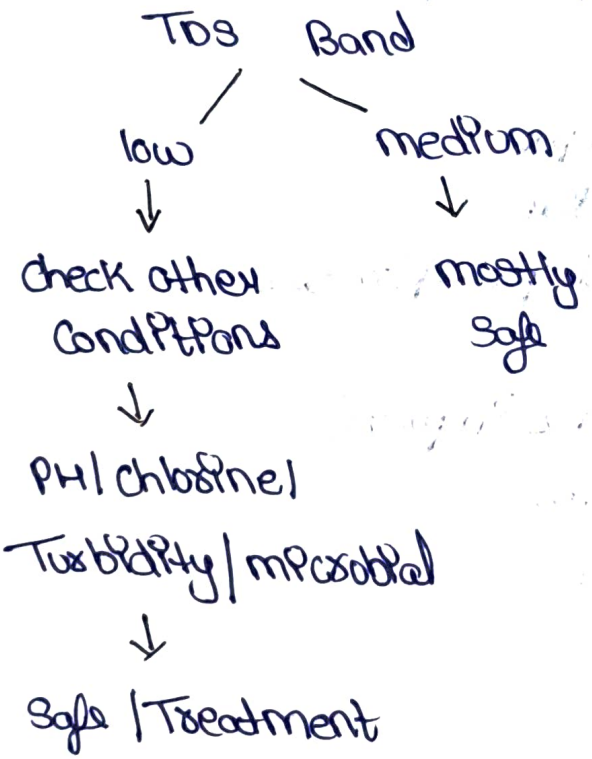
Decision Tree learning

The Decision Tree uses Entropy and Information Gain

Entropy- measures the impurity or uncertainty of a dataset

$$\text{Entropy}(S) = \sum_{i=1}^{c} P_i \log_2(P_i)$$

Decision Tree Interpretation:



Dataset for python Implementation:-

for the computational experiment a synthetic dataset

Containing 200 records is generated

The dataset Contains:

→ 200 records

→ 6 input attributes

→ 2 output classes

Complete Google Colab python Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import
train_test_split
from sklearn.compose import
Column Transformer
from sklearn.tree import
DecisionTreeClassifier
from sklearn.tree import plot_tree
export_text
from sklearn.metrics import (
    confusion_matrix,
    accuracy_score,
    precision_score
    recall_score,
    f1_score,
    classification_report
)
```

```
# 1. Generate Synthetic Dataset
rng = np.random.default_rng(42)
n = 200
rows = []
labels = []
for i in range(n):
 if i < 100:
  row = [
  "Neutral"
  rng.choice(["low"])
  rng.choice(["low", "medium"]),
  "Normal"
  "Absent"
  rng.choice([
  "marginal"
 "Ground water",
  "Borewell",
  "Surface"
 ])
]
  Safe_condition = (
  row[0] == "Neutral"
 and row[1] == "low"
 and row[2] in ["low", "medium"]
```

```
    and row[3] == "Normal"
    and row[4] == "Absent"
    )
    while safe condition
    labels = np.array(labels)
    flip = np.random(n) < 0.04
     labels = np.where(
     flip
     np.where(
       labels == "Safe for use",
         "Need Treatment",
         "Safe for use"
  ),
  ) labels
```

2. Create Data frame

```
    df = pd.DataFrame(
          focus
        Columns = [
            "pH Band":
            "TurbidPty",
            "TDS Band",
           "Residual chlorine",
           "Microbial Indicates",
           "Source Type.
df["Class"] = labels
```

```
        Print ("Dataset Shape")
        Print (df.shape)
X = df.drop (columns = ["class"])
y = df ["class"]
Categorical_features = X.columns.tolist()
X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size = 0.20,
        random_state = 42
)
Preprocess = ColumnTransformer ([
    ('cat', OneHotEncoder (handle_unknown = "ignore"),
X.columns)
model = Pipeline ([
    ('preprocess', preprocess),
    ("tree", DecisionTreeClassifier(
        criterion = "entropy",
max_depth = 6, random_state = 42)]
model.fit (X_train, X_test)
y_pred = model.predict (y_test)
Print ("Confusion matrix:\n", Confusion_matrix (y_test, y_pred))
Print ("Accuracy:", accuracy_score (y_test, y_pred))
Print ("precision:", Precision_score (y_test, y_pred, pos_label = "yes")
Print ("Recall:", recall_score (y_test, y_pred, pos_label = "yes")
Print ("F1-score:", f1_score (y_test, y_Pred, Pos_label = "yes")).
```

Expected Model Results:-

Metric	Result
Accuracy	0.9250
Precision	0.9048
Recall	0.9500
f1-Score	0.9268

Confusion matrix

$$\begin{bmatrix} 19 & 1 \\ 2 & 18 \end{bmatrix}$$

Accuracy:-

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$TP = 19,\ TN = 18,\ FP = 2,\ FN = 1$

$$\text{Accuracy} = \frac{19+18}{19+18+2+1} = \frac{37}{40}$$

$$\text{Accuracy} = 0.925 = 92.5\%$$

Precision:

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{19}{19+2} = \frac{19}{21} = 90.48\%$$

Recall:

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{19}{19+1} = \frac{19}{20} = 95\%$$

f1-Score:

$$f_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 0.9268 = 92.68\%.$$

Clean water and Sanitation:

The model demonstrates how machine learning can assist in preliminary water-quality screening.

→ faster preliminary screening
→ Identification of sample that may need further testing
→ Support for community level monitoring
→ Improve accuracy of water quality.

SDG 3- Good health and well-Being

→ unsafe water can contribute to health problems
→ A screening system can help identify samples that require further investigation before consumption.
→ Support preventive health practices by helping prioritize potentially problematic water samples.

False safe and false alert RISKs

false safe:

A safe prediction occurs when

Actual - needs Treatment

Predicted = safe for use

THIS can be dangerous because a user may incorrectly assume the water is safe

false Alert:-

Actual = Safe for use

Predicted = Needs Treatment

→ Unnecessary treatment → additional cost,
→ Inconvenience → Unnecessary Investigation

Limitations:-

→ Dataset is Synthetic
→ only a small no of features are used
→ water quality can change with location and season
→ Categorial bands may lose detailed Information
→ model accuracy may differ on real-world data.

Conclusion:-

This project shows how machine learning can be used to screen water quality Candidate elimination helps identify the concept boundary while the Decision Tree uses Entropy and classifies water samples are Safe or needs treatment It is simple and easy to understand, but the dataset is synthetic may not represent all real-world conditions overall the project demonstrates the useful application of machine learning for clean & Safe water screening.